was reused. The route every one of these bodies of data takes, from the run that produced it to the tensors a model is fitted on, is shown in Figure 4.1, and the sections below follow it in order.

4.1 Dataset Objectives and Design Requirements

Before any run was collected I fixed four requirements that the data had to satisfy jointly They are stated explicitly, because the rest of the chapter is essentially the record of discovering that no single collection strategy meets all four.

R1: realistic operating distribution. The learned models are deployed inside a tracking controller, so they must be accurate in the states that controller visits: coordinated multi-axis motion, gravity-loaded poses, moderate speeds. Data collected far from that distribution would be accurate where it does not matter.

R2: causal identifiability of the valve. A forward model is only useful to a policy if it responds to the command. This requires that the valve signal in the training data carry information that the state history does not already carry, which is a property of the *experiment*, not of the network [Ljung 99, Isermann 11].

R3: variation across a plant family. Robust training and, later, latent adaptation both need many plants rather than one. The variation has to be physically meaningful it has to demonstrably change the measured response, and it has to be recorded per run, because the privileged latent of Section 5.5 is computed from those recorded factors.

R4: clean separation. Train, validation, and test material must be separable at the level of complete runs, so that no transition window straddles a split boundary and no run contributes to two splits.

R1 and R2 pull against each other, and that tension is the organising idea of this chapter A controller that produces realistic motion computes its command from the measured state, which makes command and state history redundant and destroys identifiability (Section 2.4.2). An experiment that guarantees identifiability has to break the feedback path, which immediately makes the motion unrealistic. Rather than compromise on one of them, I collected both: a closed-loop campaign that satisfies R1 and R3, and an open-loop campaign that satisfies R2. Section 4.5 explains how the two are used together without letting either one dominate.

4.2 Randomized Parameter Space of the Simscape Campaign

This section documents the randomization design of the Simscape curriculum. Its status must be stated before its content, because the design was drawn up under an earlier plan in which randomized *Simscape* data would be the substrate for adaptation, and that plan is superseded. No model reported in this thesis is trained on this curriculum. It is retained